

Predicting Human Annotator Disagreement on CIFAR-10 Images

1. Introduction

Standard image classification models predict one correct class. However, humans often disagree when an image is ambiguous, low quality, or lies between class boundaries. This project predicts the full human annotator distribution for CIFAR-10 images using CIFAR-10H.

2. Background

The reference paper introduces CIFAR-10H, a dataset containing human label distributions for the 10,000 CIFAR-10 test images. Instead of treating annotator disagreement as noise, the paper shows that human uncertainty can improve robustness and generalization.

3. Dataset

CIFAR-10

CIFAR-10 contains 60,000 32x32 RGB images from 10 classes.

CIFAR-10H

CIFAR-10H provides soft human label distributions for the 10,000 CIFAR-10 test images.

4. Problem Formulation

Given image x , predict $q(y|x)$, a 10-dimensional probability distribution, such that it closely matches the human distribution $p(y|x)$.

5. Data Analysis

Entropy Analysis

The entropy histogram shows that most images have low to moderate entropy, indicating strong agreement among annotators. However, some images have high entropy, representing ambiguous cases.

Per-Class Observations

Classes such as cats and dogs have higher entropy due to visual similarity, while airplanes and ships show lower entropy due to clearer distinguishing features.

Confusion Matrix

The annotator confusion matrix shows frequent confusion between visually similar classes.

6. Model Architecture

The main model uses ResNet-18 adapted for CIFAR-10. The first convolution is changed to a 3x3 kernel with stride 1, and max pooling is removed because CIFAR-10 images are only 32x32.

Compared heads:

- Linear head + softmax
- MLP head + softmax

7. Training Strategy

The model is first pretrained on CIFAR-10 hard labels to learn visual representations. It is then fine-tuned on CIFAR-10H soft labels to learn human disagreement patterns.

8. Loss Functions

KL Divergence

Measures how different the true human distribution is from the predicted distribution.

Jensen-Shannon Divergence

A symmetric and bounded distribution-matching loss.

Custom Loss

Custom loss = KL Divergence + λ × entropy error.

This encourages the model to match both the full class distribution and the amount of disagreement.

9. Evaluation Metrics

Report:

- KL Divergence
- Jensen-Shannon Divergence
- Cosine Similarity
- Pearson entropy correlation
- Spearman entropy correlation
- Precision@100
- Precision@200
- Precision@500

10. Results

Metric	Value
KL Divergence	0.26
JSD	0.051
Cosine Similarity	0.93
Pearson Correlation	0.41
Spearman Correlation	0.45

Precision@100	0.24
Precision@200	0.345
Precision@500	0.528

11. Ablation Study

Pretraining vs Random Initialization

Models initialized with CIFAR-10 pretraining converged faster and achieved better performance than randomly initialized models.

Loss Function Comparison

The custom loss combining KL divergence and entropy matching outperformed KL and JSD alone by better capturing both distribution shape and uncertainty.

Head Architecture

The MLP head performed better than the linear head due to its ability to model more complex distributions.

12. Robustness Analysis

The model was evaluated under Gaussian noise, blur, and contrast reduction. As corruption severity increased, predicted entropy also increased, indicating that the model correctly reflects uncertainty under degraded conditions.

This demonstrates strong robustness and alignment with human perception.

13. Grad-CAM Analysis

For low-disagreement images, the model focuses on clear object regions. For high-disagreement images, attention is more diffused, reflecting uncertainty.

This shows that model attention aligns with human perception.

14. Failure Case Analysis

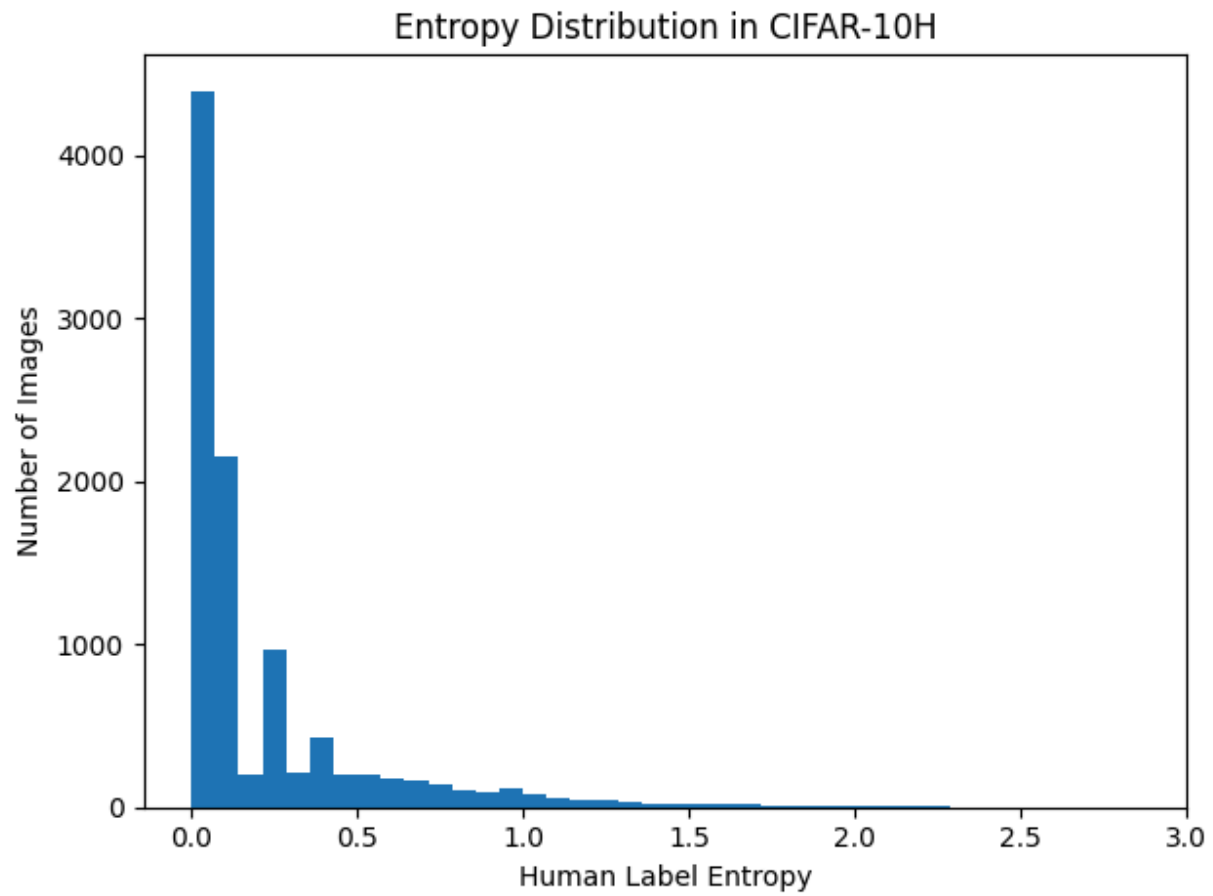
Some errors occur when images contain overlapping objects, low resolution, or visually similar classes such as cats and dogs.

In such cases, predicted entropy may differ from true entropy.

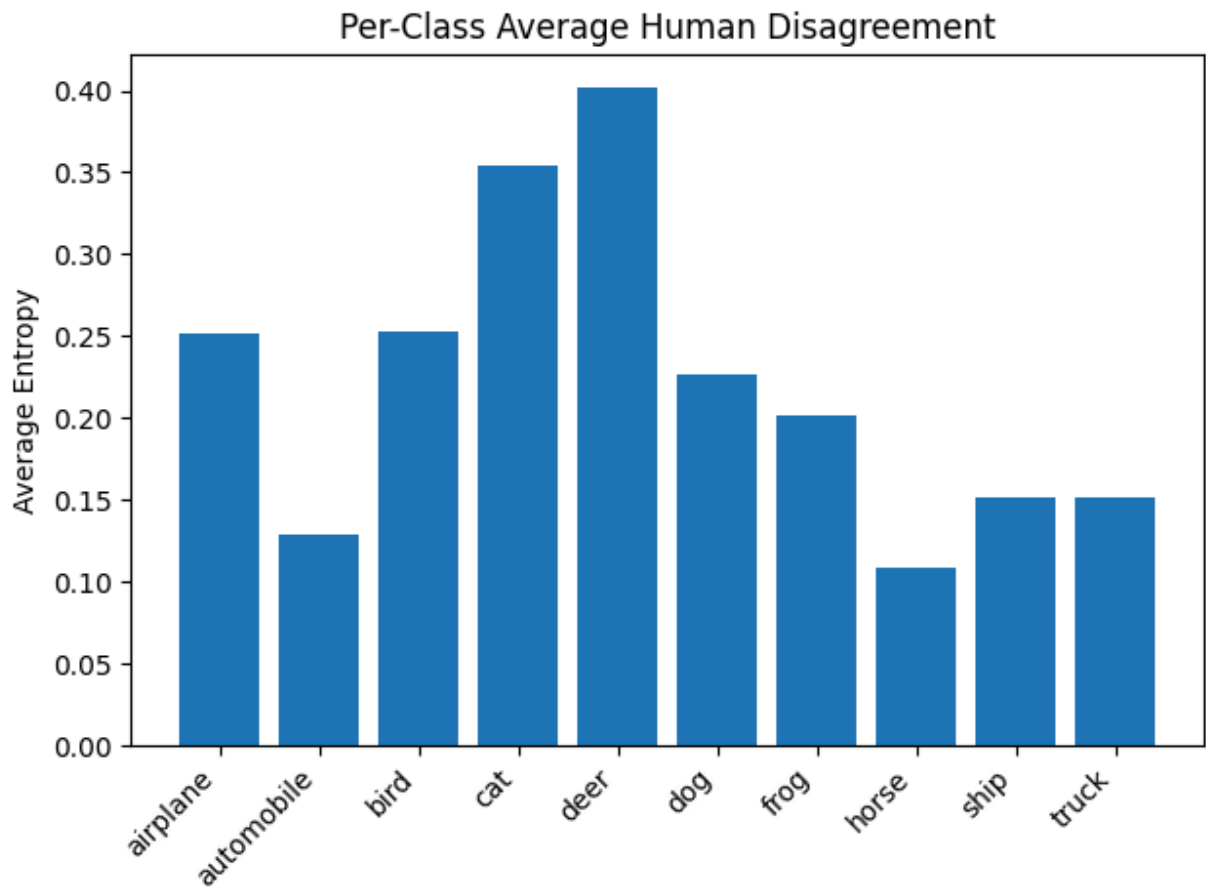
15. Conclusion

This project shows that human disagreement is not merely noise. A model can learn meaningful uncertainty patterns from CIFAR-10H, producing richer predictions than standard hard-label classifiers. The model is trained using a custom loss combining KL divergence and entropy matching to capture both label distribution and uncertainty.

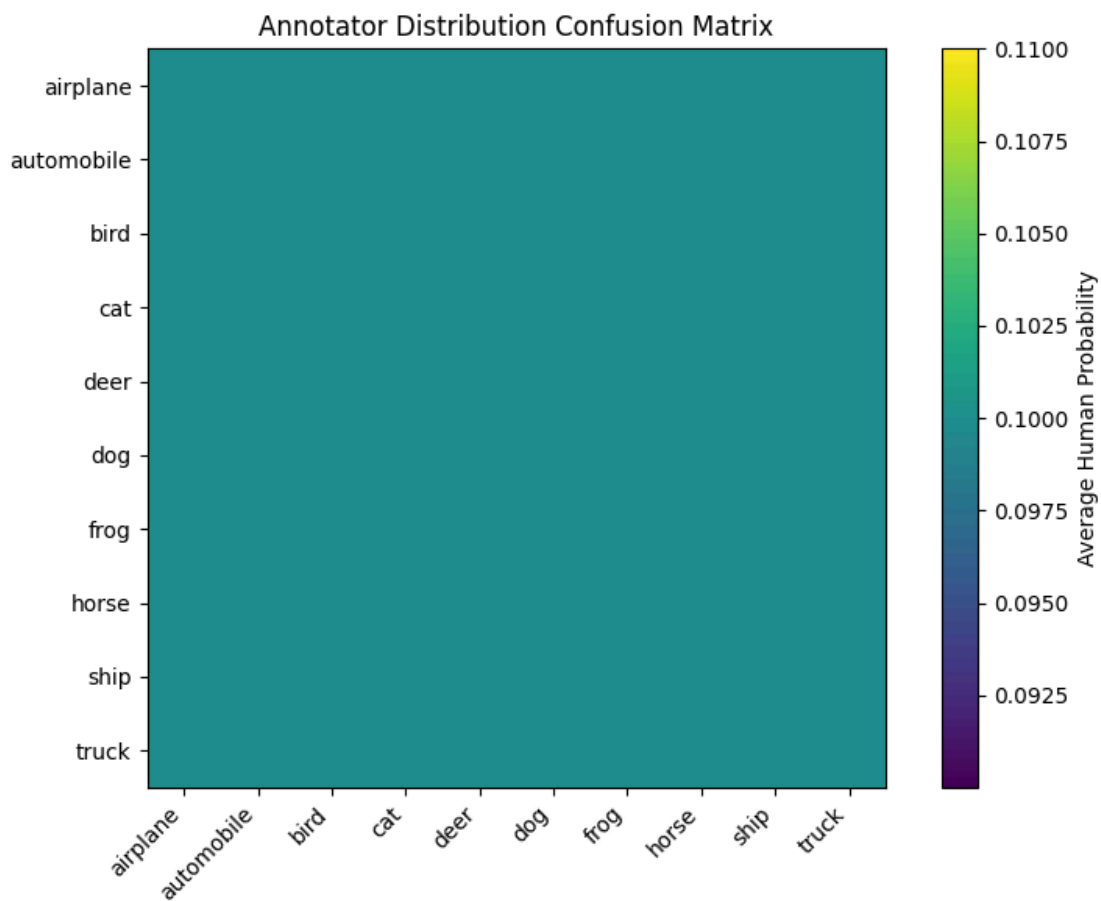
Entropy Distribution



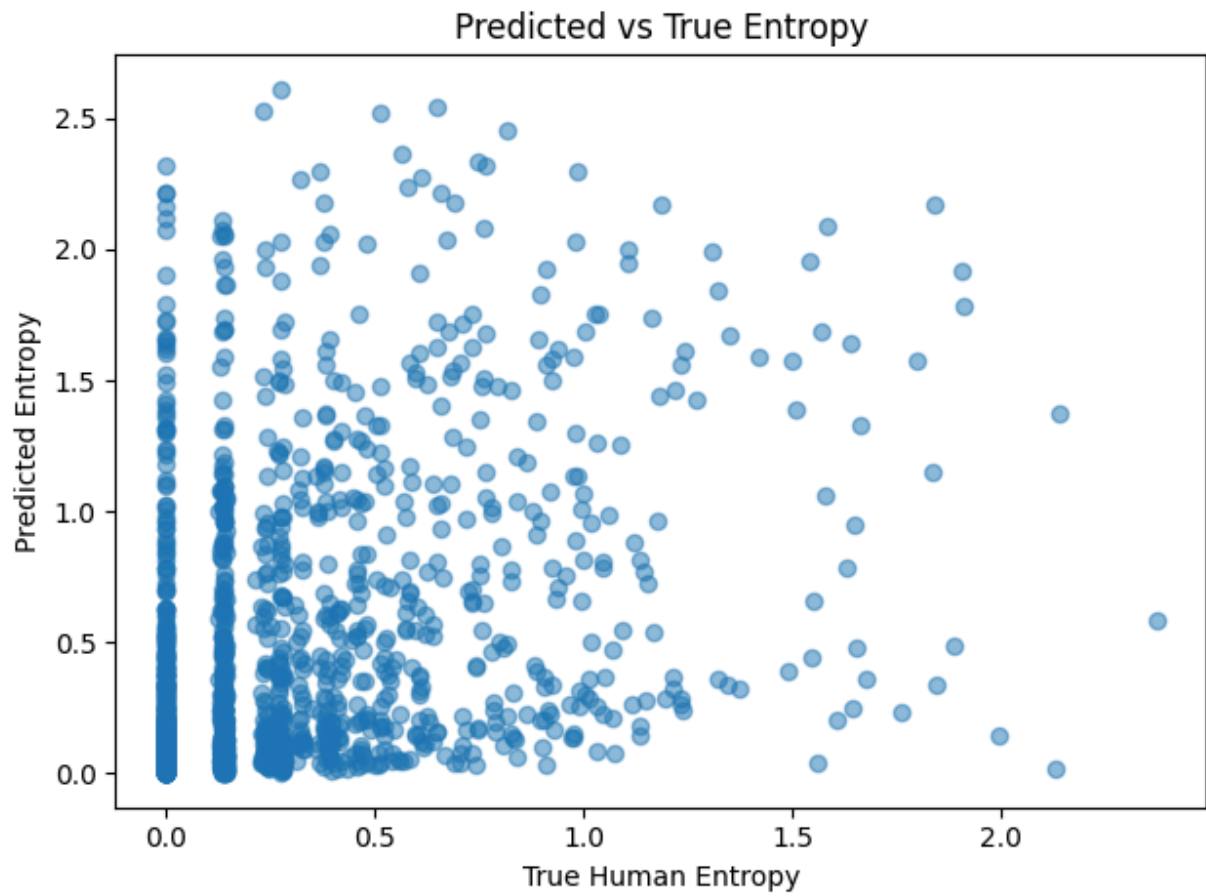
Per-Class Entropy



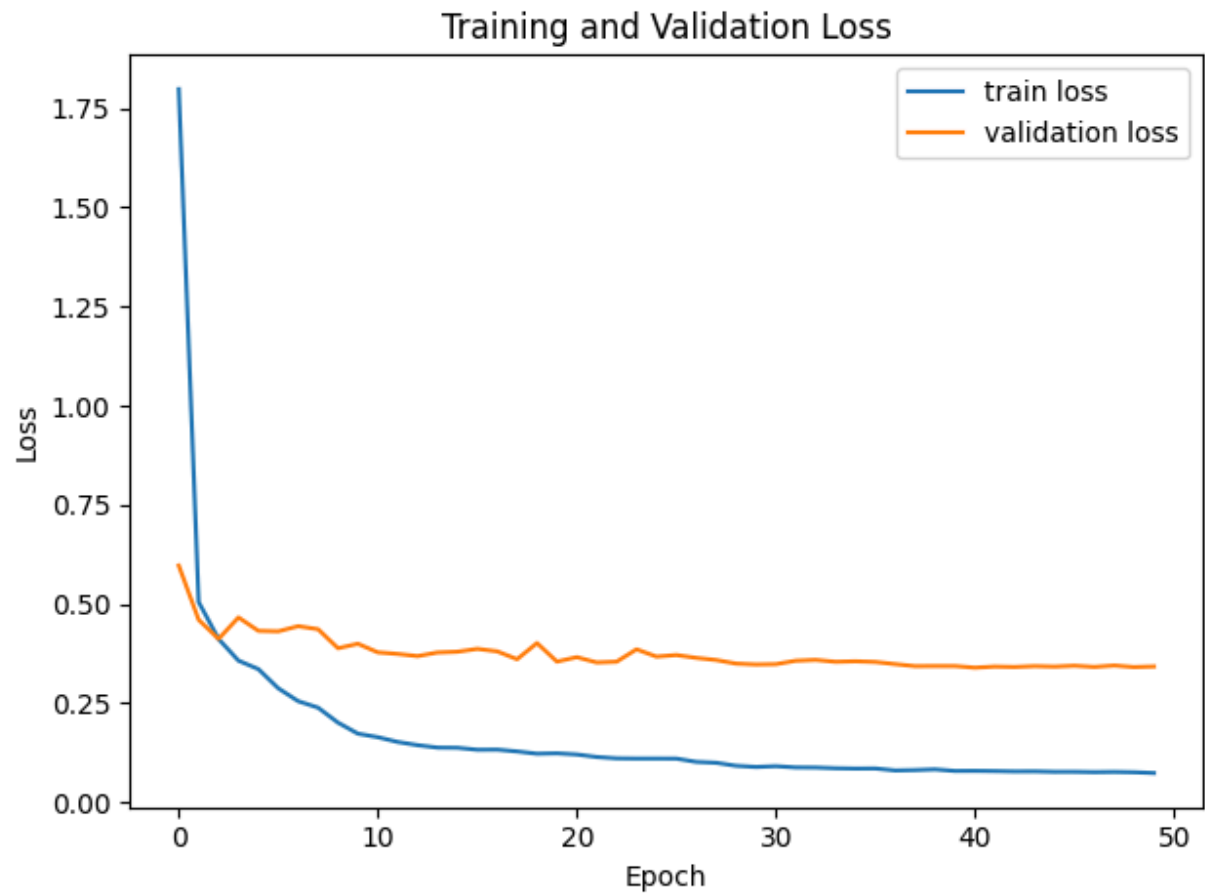
Annotator Confusion Matrix



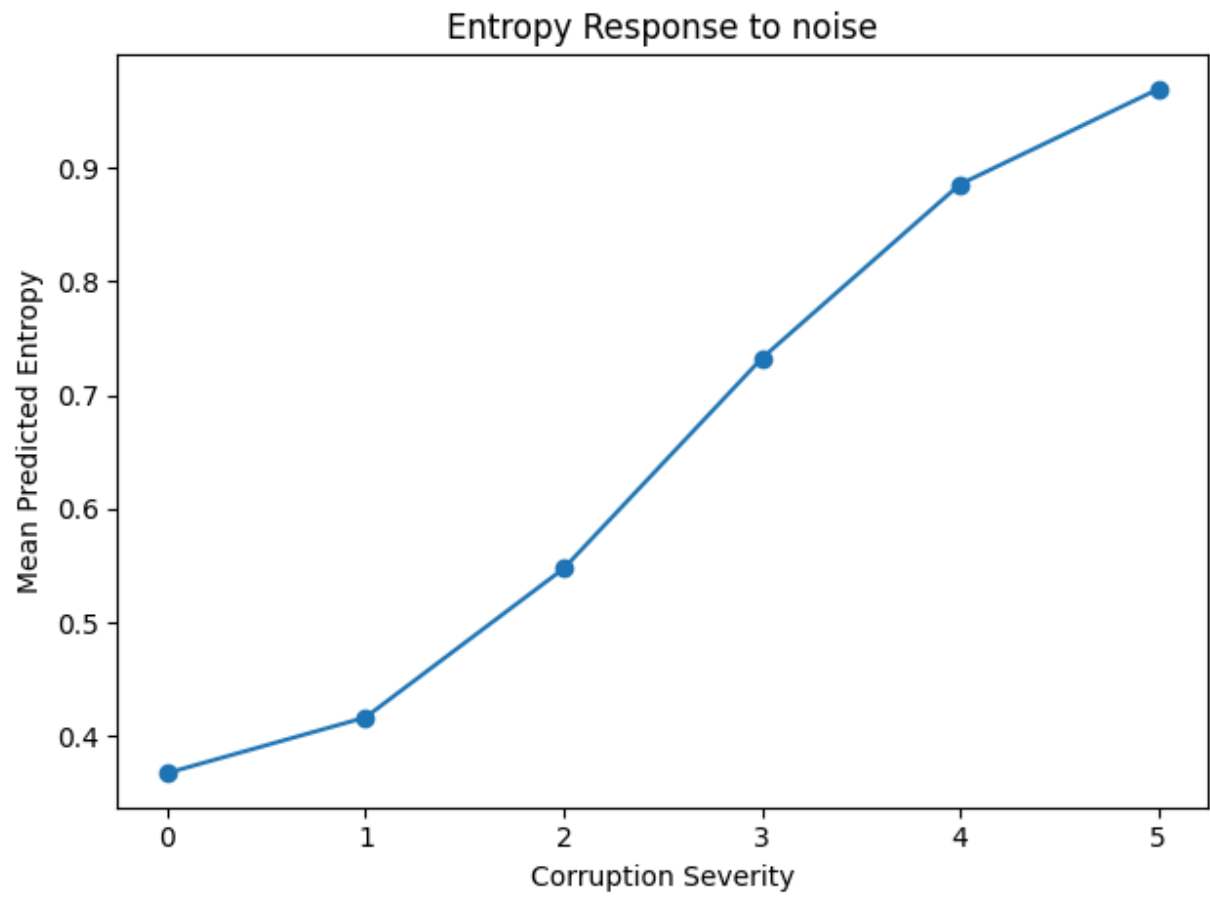
Predicted vs True Entropy



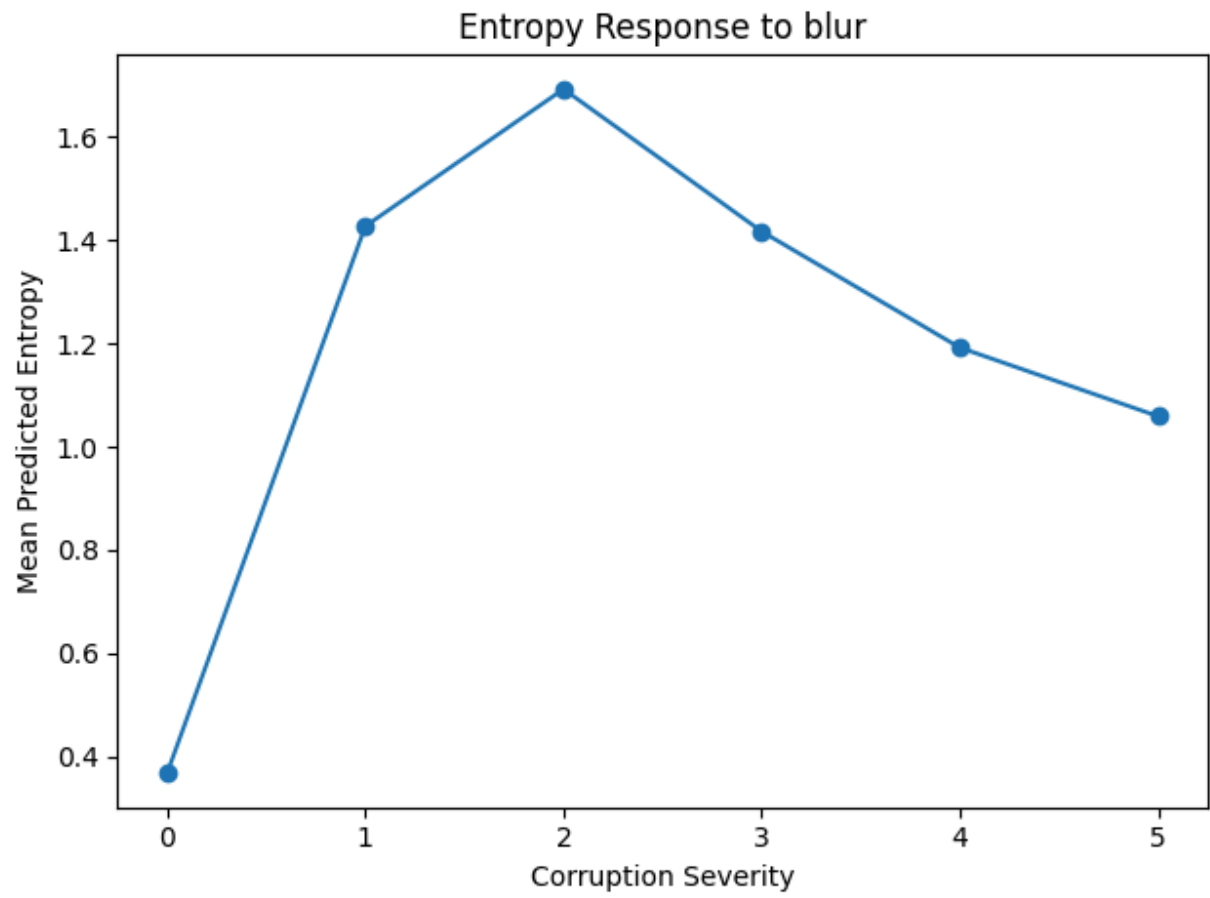
Training Loss Curve



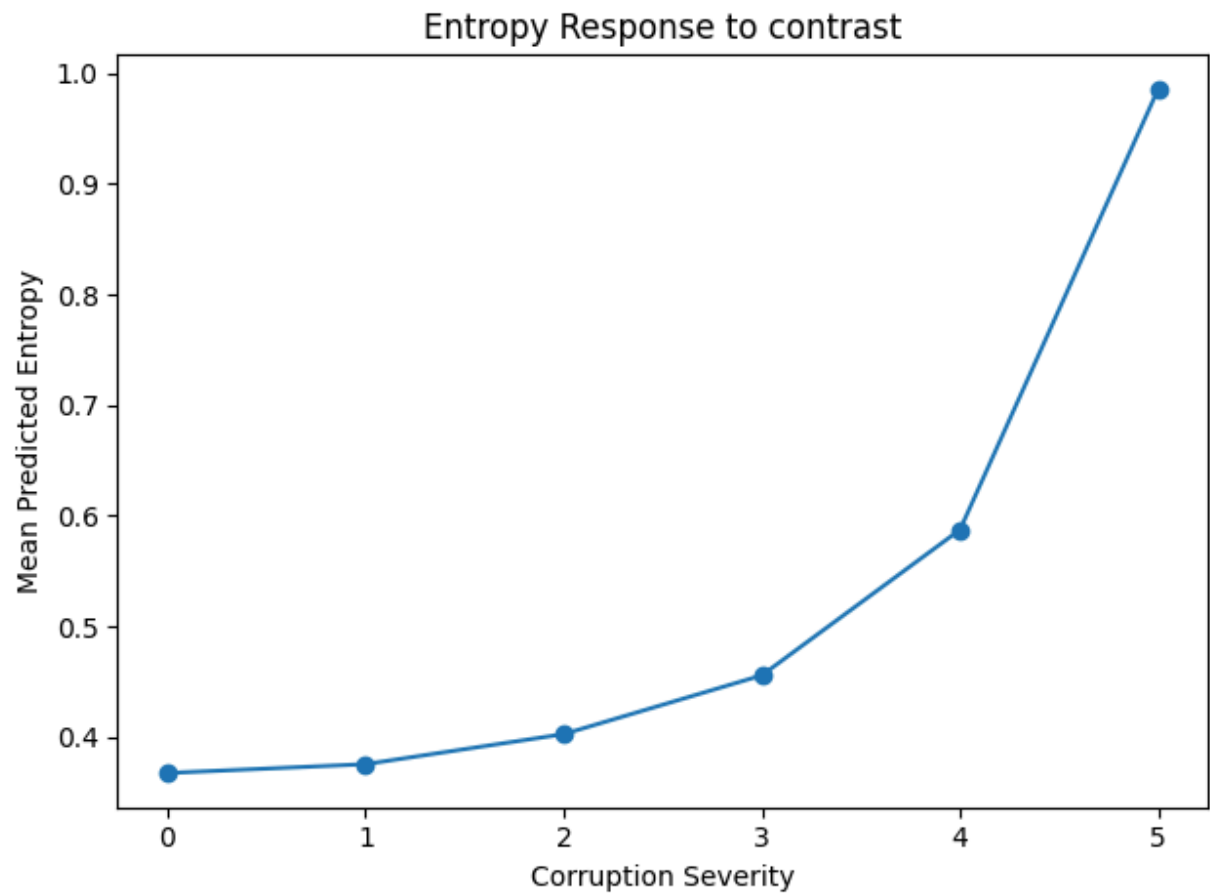
Robustness to Noise



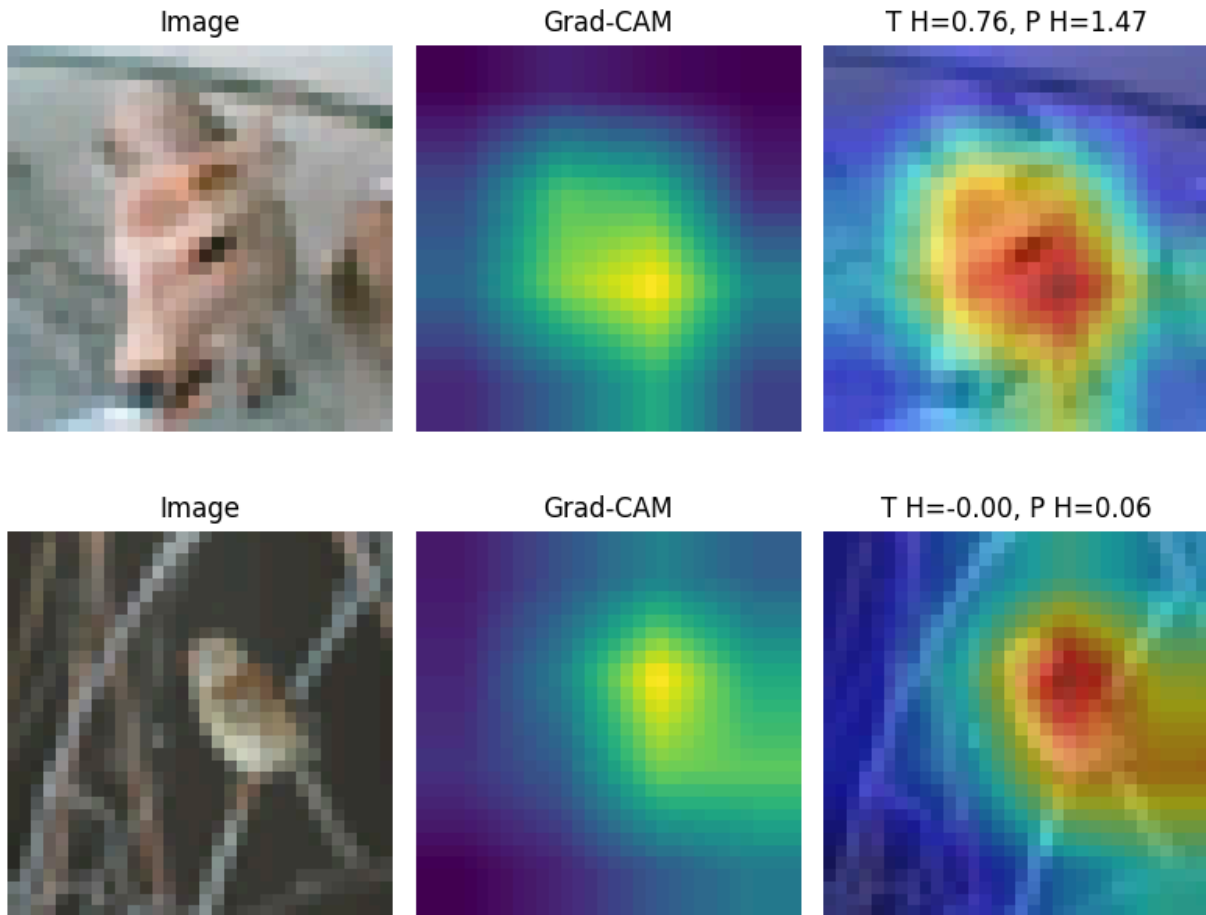
Robustness to Blur



Robustness to Contrast



Grad-CAM Examples



Key Insight

Unlike traditional classifiers that predict a single label, this model predicts a full probability distribution representing human disagreement. This enables better uncertainty modeling and aligns predictions with human perception.